Evaluating Retrieved Episodic and Personalized Semantic Memory in Small Local Language Models

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***Abstract*—Episodic memory recalls a specific past event; semantic memory holds generalized knowledge abstracted from many events [1]. For a language-model assistant these are two different jobs: reconstruct what happened, and apply a stable model of the user. This project asks how retrieved episodic event memory and personalized semantic profile memory affect the answer quality of small, local, quantized models. Memory is not varied as an experimental condition here; it is always retrieved and supplied. The operational question is where quality breaks down once memory is in context: retrieval or generation, and how that depends on model size. We assemble a benchmark-first pipeline over two benchmarks, EPBench for episodic events [7] and LaMP-QA for personalized QA [12], retrieve with a 300M EmbeddingGemma model [9], and generate with two Gemma 4 models, E2B and E4B [10], served locally through llama.cpp. Retrieval is near-ceiling (hit@10 = 1.000, recall@10 = 0.996 over 276 EPBench queries), so generation is the differentiating factor. On personalized-semantic answering the larger E4B model is clearly better (LaMP-QA 0.526 vs. 0.477; paired 95% CI $[+0.013, +0.086]$); on episodic recall its point estimate is higher (EPBench 0.691 vs. 0.660) but the paired difference is within sampling noise. Reasoning on the ten hardest items per benchmark recovers them, most for E4B. The bottleneck for a small local memory assistant is not retrieval but generation, and personalization is where the model-size effect is real.**

I. Introduction

A small language model running on a personal device is private, cheap, and always available, but it carries a hard constraint: a bounded context window. That constraint has a direct cognitive analogue in the limited capacity of human working memory, which holds only a few items at once [14]. External memory is the standard workaround. It stores past interactions and user information, embeds them, and retrieves the most relevant fragments into the prompt. The design question is not whether to add memory but what kind of memory to retrieve, and whether a small model can actually use it once it is there.

Cognitive science gives a clean way to split the problem. Tulving separates episodic memory, which records specific events bound to a time and context, from semantic memory, which holds generalized knowledge abstracted across experiences [1]. The two map onto two different assistant capabilities: recalling what happened (episodic) and applying a stable profile of the user (semantic). They are not interchangeable. Recent work treats human-like episodic memory in language models as a research target in its own right [5], while a separate line models durable user-specific knowledge as a persona or profile memory [6]. Treating the whole pipeline as a cognitive model is justified by the computational-representational view of mind: cognition is mental representations plus procedures that operate on them [2].

The project started as something else. The pitch proposed a synthetic productivity assistant comparing no-memory, episodic, semantic, and combined conditions. During implementation the scope narrowed, and the commit history records the decision points directly: one commit adds the personalization pipeline "without generated dataset," dropping synthetic data for established benchmarks, and a later commit removes an exploratory third benchmark ("No QASPER"). The assistant remains the motivation; the delivered artifact is a benchmark-first evaluation. The motivating question is broad:

> *How does retrieved episodic event memory and personalized semantic profile memory affect small language-model answer quality on benchmarked memory tasks?*

This study does not vary memory as an experimental condition. Retrieved memory is always supplied, so the study cannot isolate the effect of adding episodic or semantic memory against a no-memory baseline. It answers a sharper operational question instead: once both memory types are retrieved and placed in context, where does answer quality break down for a small local model, in retrieval or in generation, and how does that depend on model size? The two questions connect. A memory-condition comparison would still have to locate the bottleneck this study measures, and the answer decides whether more retrieval or a better generator is the productive next step.

The contribution is an evaluation, not a new memory architecture. We (i) map the episodic/semantic split onto two benchmarks with complementary grading; (ii) hold retrieval fixed and report retrieval quality and generation quality separately, with paired confidence intervals on the model-size effect; and (iii) compare two model sizes and a targeted reasoning condition on identically retrieved context.

II. Cognitive-Science Background

Episodic and semantic memory. Tulving characterized episodic memory as the store of temporally dated episodes and their spatio-temporal relations, and semantic memory as

organized knowledge of facts and concepts abstracted away from the events that produced it [1]. The difference matters for retrieval: an episodic query ("where was the meeting?") needs one specific event with its bound details, while a semantic-profile query ("recommend a restaurant for me") needs generalized knowledge of the user's preferences. Our two benchmarks operationalize exactly this split: one asks about specific events, the other conditions on a user profile.

A computational reading. The computational-representational understanding of mind treats thinking as representations plus procedures that manipulate them [2]. That maps onto the pipeline directly. Stored memory records are the representations; embedding, retrieval, and prompt construction are the procedures; the bounded context window is a working-memory buffer; and the generated answer is the behavior we score. The Standard Model of the Mind, a consensus architecture across cognitive science and AI, casts long-term memory as distinct declarative stores that explicitly separate episodic and semantic memory, accessed by cue-based associative retrieval into that buffer [3]. Vector similarity search is a simplified version of that associative access, nothing more.

Why a small top-k. Working memory is capacity-limited; Miller's estimate of roughly seven chunks is the canonical statement that only a few items can be held and manipulated at once [14]. A language model inherits an analogous limit from its context budget and from the cost of long prompts on-device, which is why the system retrieves a small top-k rather than the whole corpus. The consequence sets up the evaluation: retrieval quality, meaning whether the right memory reached the few available slots, is a separate question from whether the model then uses those slots well.

III. Related Work

Episodic memory for LLMs. A growing literature treats episodic memory as a capability to engineer into language models and to test directly [5], and purpose-built benchmarks generate synthetic event histories to check whether a model can answer questions about specific past episodes [7]. We adopt one such benchmark instead of proposing a new episodic mechanism.

Personalized and semantic-profile memory. The LaMP family frames personalization as conditioning generation on a user's history, retrieving per-user profile items to tailor an answer [8]; its long-form QA successor supplies the personalized questions we evaluate [12]. Related work models durable user-specific knowledge as a temporally-aware persona or semantic memory [6].

Memory-augmented systems. PRIME gives a model an explicit dual episodic-plus-semantic memory [4], and MemGPT manages a tiered memory hierarchy, paging information in and out of the context window like an operating system [16]. These build memory architectures. We do the opposite: hold the retrieval mechanism fixed and measure the effect of retrieved memory on small models.

Retrieval. Dense passage retrieval established learned bi-encoder embeddings as a strong alternative to sparse lexical matching for open-domain QA [17], and retrieval-augmented generation paired a dense retriever with a generator to condition output on retrieved text [15]. Applied work pairs semantic search with in-context injection to improve retrieval without fine-tuning [11]. Our pipeline is one instance of this retrieve-then-generate pattern.

IV. Model/Tool Design

System overview. The system is a set of standalone scripts wired into one append-only pipeline: normalize, validate, run (retrieve and generate), then grade/judge (Fig. 1). Retrieval and generation run against two separate local llama.cpp servers, an embedding server on port 8082 and a chat server on port 8081, so the two stages stay decoupled. Each run writes `results.jsonl`, `metrics.json`, and `manifest.json`. The manifest records the true chat model, and it is authoritative: the runner scripts are reused across model sizes, so file and field names cannot be trusted to identify the model.

Benchmarks. For episodic memory we use EPBench [7], which builds synthetic event books and asks questions about specific events. The processed set is 276 queries over a 19-document corpus with 378 relevance judgments, across six query types: Times, Entities, Spaces, Event contents, Full event details, and Other entities. For personalized semantic memory we use LaMP-QA [12], a personalized long-form QA benchmark descended from LaMP [8]. We evaluate a 100-question sample; each question carries a user profile averaging roughly 114 items.

Retrieval. One EmbeddingGemma-300M model [9], served as an 8-bit quantized GGUF, embeds both queries and memory candidates. Retrieval ranks candidates by cosine similarity, equivalently a dot product over normalized embeddings, and returns the top-k ($k = 10$ for EPBench, 5 to 8 for LaMP-QA) from the full-corpus pool. This is a dense-retrieval [17], RAG-style [15] front-end whose retrieved items are injected into the prompt as in-context evidence [11]. Because a single embedder and a fixed k are shared across generators, retrieval is identical no matter which generator runs. That is what lets us attribute any downstream difference to generation.

Generation. Two Gemma 4 generators [10], E2B and E4B, are served as 4-bit quantized GGUFs with a 128K context window, at temperature 0 for reproducibility. Model size is the primary comparison. One reproducibility caveat matters: the LaMP-QA runner stores every generated answer under a field named `e2b_answer` even for E4B runs, so the model must be read from `manifest.json`, not from the field name.

Two axes, plus a reasoning probe. We keep two axes separate: retrieval versus generation, and E2B versus E4B. Separately, we re-run the ten lowest-scoring items per benchmark with the model's reasoning ("thinking") mode enabled. This is an error-analysis probe on the hard tail, not a full thinking condition over the whole distribution. The results keep that distinction.

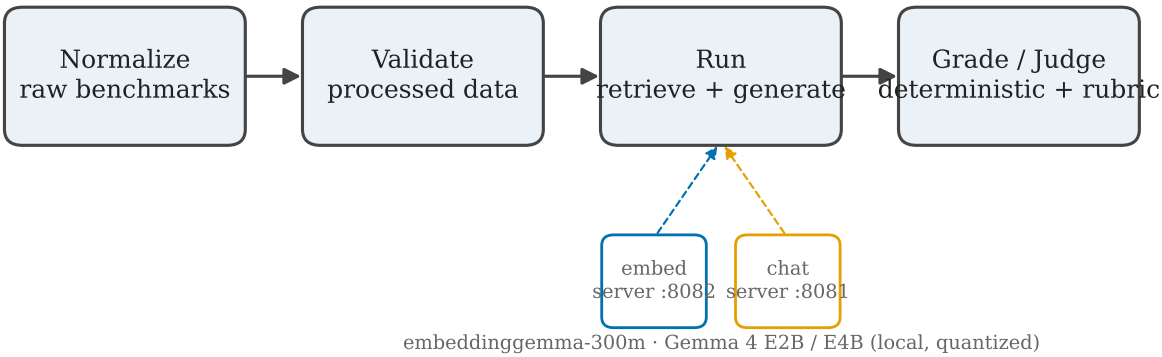


Fig. 1. The standalone-script evaluation pipeline. Retrieval and generation run against two local `llama-server` instances (embedding :8082, chat :8081); EPBench is graded deterministically and LaMP-QA by a rubric judge.

Grading and uncertainty. EPBench is graded deterministically. The primary score is the mean of per-field F1 over extracted anchors, dates, locations, and persons, combined with a semantic-cosine term computed with the same embedder. LaMP-QA has no reference answer, so it is judged against its rubric of expected aspects: each aspect is scored covered, partial, or missed, and the result is normalized to [0,1]. Using a strong model as the judge is supported by evidence that LLM judges track human preferences on open-ended answers, but the same work documents position, verbosity, and self-enhancement biases [13]. We reduce these with a fixed rubric and temperature 0, and we treat the deterministic EPBench score as the more trustworthy headline metric because it does not depend on a judge at all. For the model-size comparison we report 95% bootstrap confidence intervals on the paired per-item difference (E4B minus E2B) over the shared items, resampled 20,000 times.

V. Results

Retrieval is near-ceiling. Over the 276 EPBench queries the shared retriever reaches hit@1 = 0.888, hit@3 = 0.971, hit@5 = 0.993, **hit@10 = 1.000**, recall@10 = 0.996, MRR = 0.931, and nDCG@10 = 0.943 (Fig. 2). Because the embedder and top-k are shared, these numbers are identical for the E2B and E4B pipelines. This one result organizes the rest: the correct memory is almost always in the retrieved context, so differences in answer quality come from generation, not from what was retrieved.

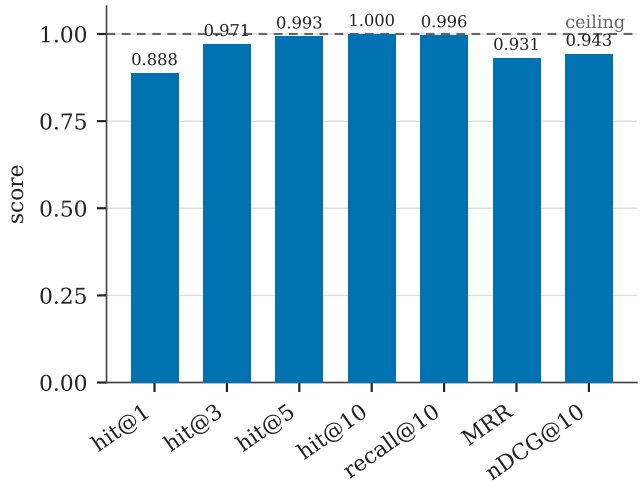


Fig. 2. Retrieval quality on EPBench ($n = 276$). hit@10 reaches 1.0 and recall@10 0.996; retrieval is near-ceiling and identical across models, so generation is the differentiator.

EPBench generation. On the deterministic primary score, E4B's point estimate leads E2B, **0.691 vs. 0.660** (mean paired difference $+0.032$). The paired 95% bootstrap CI is $[-0.008, +0.071]$ and includes zero, so on 276 queries the two models are not statistically distinguishable on EPBench: E4B wins on 35% of queries, ties on 39%, and loses on 26%. The point-estimate gain concentrates in concrete, entity-like fields (location F1 0.356 to 0.418, anchor F1 0.437 to 0.472), while E2B is marginally higher on the semantic-cosine term (0.434 vs. 0.430). Figure 3 shows the per-metric 95% intervals, and the E2B and E4B intervals overlap on every component, including the primary score. By query type (Fig. 4), both models score highest on Times and Entities and lowest on Event contents (0.399 for E2B, 0.359 for E4B), the one category where the larger model does not help, because open-ended event descriptions resist an extraction-based grader. E4B's clearest query-type gain is on Spaces, which rises from 0.62 to 0.79.

LaMP-QA generation. On the rubric score, E4B leads, **0.526 vs. 0.477** (mean paired difference $+0.049$). Here the paired 95% bootstrap CI is $[+0.013, +0.086]$ and excludes zero (two-sided $p \approx 0.009$), so the personalization gain is statistically distinguishable, unlike the EPBench gap. The per-model intervals in Fig. 5(a) overlap slightly, but the paired test,

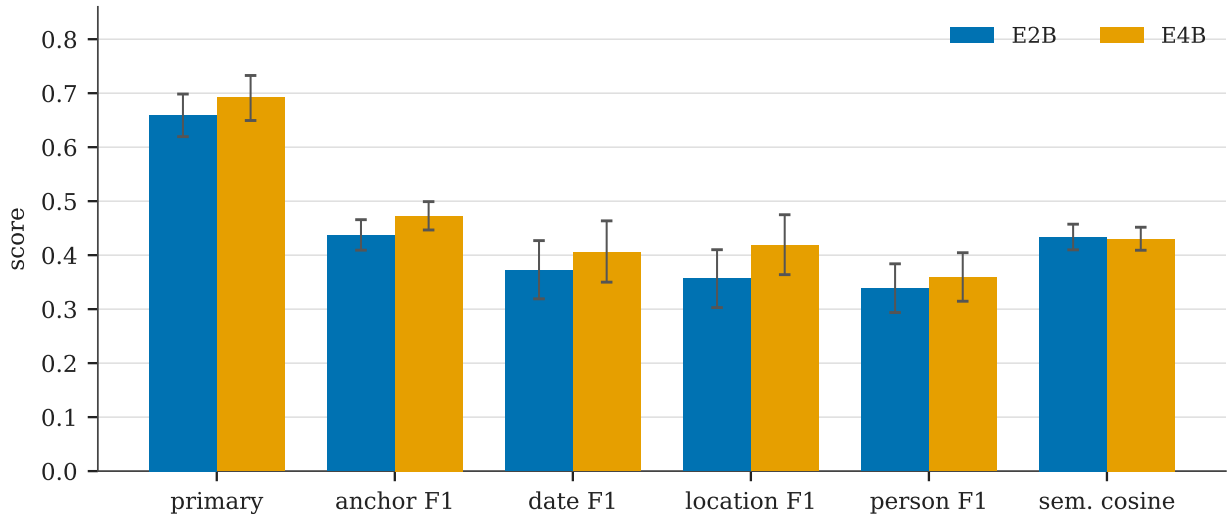


Fig. 3. EPBench generation quality by component ($n = 276$), E2B vs. E4B, with 95% bootstrap confidence intervals. The E2B and E4B intervals overlap on every component, including the primary score; E2B is marginally higher only on semantic cosine.

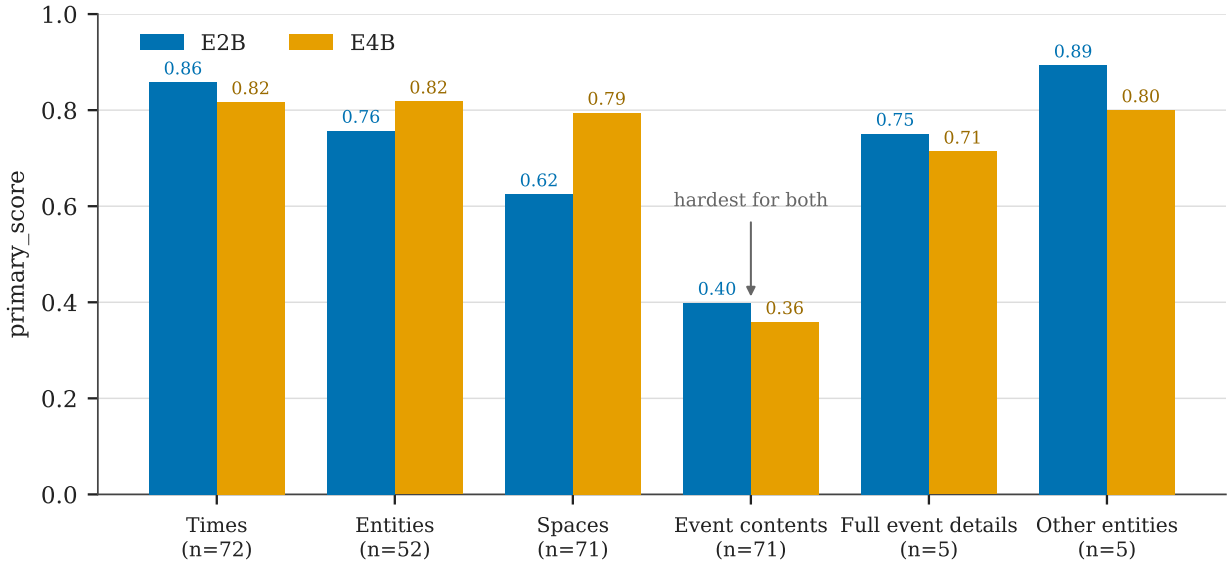


Fig. 4. EPBench primary score by query type. Both models score highest on Times and Entities and lowest on Event contents, where the larger model does not help. E4B's clearest gain over E2B is on Spaces (0.62 to 0.79).

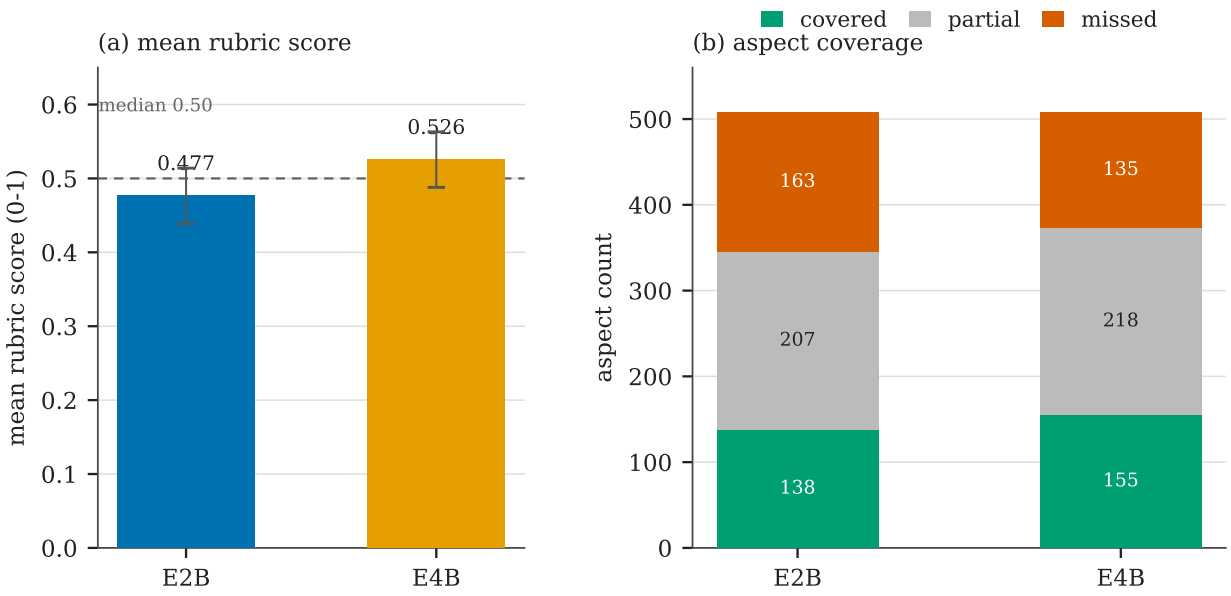


Fig. 5. LaMP-QA personalized-semantic quality ($n = 100$). Panel (a) shows mean rubric score with 95% bootstrap CIs; the per-item median stays 0.50. Panel (b) shows aspect coverage: E4B shifts aspects from missed into covered and partial. The per-model intervals in (a) overlap, but the paired difference is significant (95% CI $[+0.013, +0.086]$).

which removes between-question variance, is what establishes the effect. The per-item median is 0.500 for both models, so the improvement is a distributional shift, not a change in the typical case: E4B moves aspects out of "missed" (163 to 135) and into coverage (covered 138 to 155; partial 207 to 218). Personalization is hard for both models; neither approaches full aspect coverage.

Reasoning on the hard tail. Re-running the ten lowest-scoring items with reasoning enabled recovers them, most for the larger model (Fig. 6). On EPBench the selected items all scored exactly 0.000 in the base run, and with reasoning they reach 0.715 for E2B and 0.900 for E4B. On LaMP-QA the same items scored 0.138 (E2B) and 0.210 (E4B) at baseline and reach 0.205 and 0.579 with reasoning, a $+0.369$ gain for E4B. Two caveats bound this. Because the base run is deterministic at temperature 0, a no-reasoning re-run reproduces the baseline exactly, so the lift is attributable to reasoning rather than to resampling noise. But the items are selected as the hardest, so the magnitude of the gain does not generalize to the full distribution. One confound we do not resolve: because the items were chosen as the lowest-scoring, the lift cannot be cleanly separated from the fact that these items had the most room to improve. Isolating "reasoning helps" from "the hardest items have the most headroom" would require running reasoning on a matched sample of non-bottom items, which we did not do. This is targeted error analysis, not a thinking condition.

VI. Discussion

Separating retrieval from generation is not cosmetic. Retrieval saturates (Fig. 2), so a single blended score would hide the finding that matters: the small model's limitation is downstream of retrieval. Reporting the two axes apart is what lets us attribute the generation gap to model size. For on-device memory systems, where retrieval is usually the easy part, this separation should be the default.

What the model-size gap means. E4B's only statistically reliable advantage is on LaMP-QA personalization. On EPBench its higher point estimate concentrates in location and entity fields, but the paired difference is within noise on 276 queries, so we do not claim a dependable episodic-recall advantage from model size alone. Read through the episodic/semantic distinction [1], the robust finding is that the larger model helps most where the task is to synthesize a generalized user profile into an answer, not where it is to reproduce the bound details of a single event. Personalization is also where both models are weakest in absolute terms: medians pinned at 0.5 indicate a use gap, not a retrieval gap. The relevant profile items are retrieved, but neither small model reliably converts them into a personalized answer. That is the frontier dual-memory architectures such as PRIME [4] target.

What this says about the cognitive model. The computational-representational account from Section II gives the pipeline two stages [2]: associative retrieval that brings representations into the working-memory buffer, and procedures that operate on those representations to produce behavior. The results place the small-model bottleneck in the second stage. Retrieval, the associative-access analogue, is saturated (hit@10 = 1.000), so the representations the model needs are already in the buffer. What limits answer quality is the procedural step that reads those representations and composes an answer,

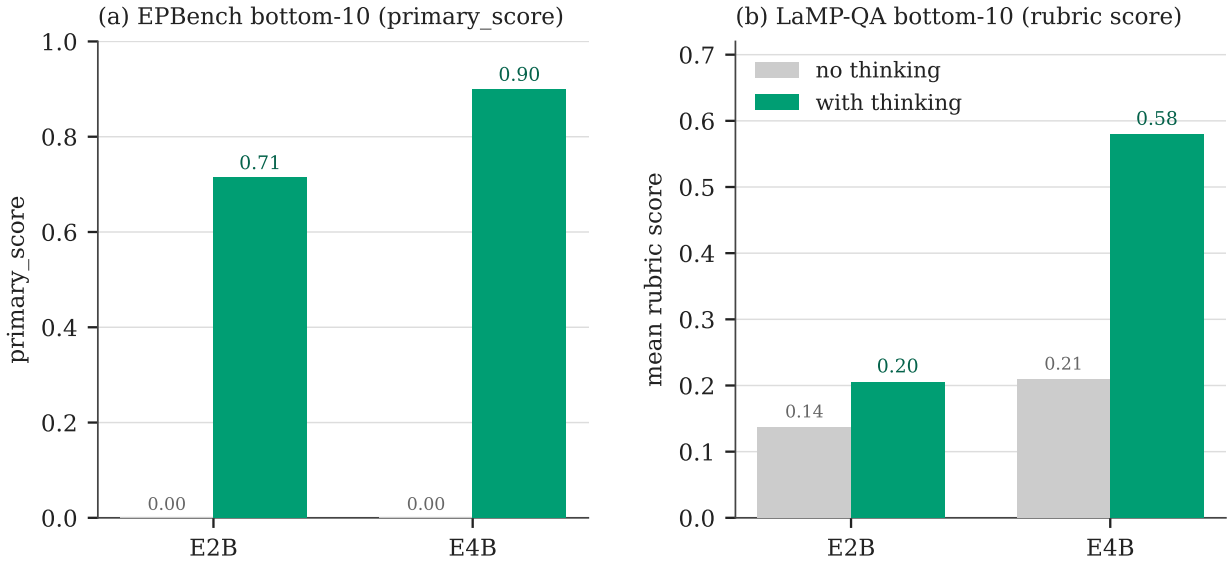


Fig. 6. Reasoning on the ten lowest-scoring items per benchmark. The EPBench items all scored 0.000 in the deterministic base run; LaMP-QA baselines are 0.138 (E2B) and 0.210 (E4B). Reasoning recovers the hardest cases, most for E4B. Selection to the hard tail means the magnitude does not generalize, so this is targeted error analysis, not a full thinking condition.

and that step is where model size and reasoning make their difference. In the terms of the Standard Model of the Mind [3], the declarative-retrieval component is effectively solved at 300M parameters, while the procedural component that uses retrieved memory is the constraint. This reframes "add more memory" as the wrong lever for a small local assistant: the relevant items are already retrieved, so the productive work is in the procedures that consume them, not in storing or fetching more.

Practical implication. For a private, on-device memory assistant, a 300M embedder already delivers near-ceiling associative recall, and a 4B-class local model answers concrete episodic questions reasonably well (about 0.69 on EPBench). The limiting factor is personalization quality, where scores sit near 0.5 and where the extra parameters of E4B produce the one improvement we can call statistically reliable. Reasoning helps the hardest cases, but at a compute cost that offsets part of the reason to run small and local in the first place.

VII. Limitations

Several limits bound these claims. The benchmarks are proxies for the motivating productivity setting, not real user tasks; there are no human participants and no interactive assistant, so we measure benchmark answer quality, not usefulness in practice. The samples are modest (276 EPBench queries, 100 LaMP-QA questions) and the model-size effects are small; on EPBench the paired difference is not significant, so that comparison is suggestive only. All models are quantized (4-bit generators, 8-bit embedder), which likely depresses absolute scores. LaMP-QA is graded by an LLM judge, an approximation with known biases [13] that we reduce but cannot remove. The reasoning sub-evaluation covers only the ten hardest items per benchmark, so it does not by itself support a claim about the full distribution. Most important: the study contains no none/episodic/semantic/combined memory-condition ablation. The two axes actually varied are model size (E2B vs. E4B) and reasoning on/off on the hard tail, and every claim here is scoped to those two axes. The larger comparison the pitch envisioned is future work, not a result of this study.

VIII. Conclusion

The bottom line is a split. For small local language models, associative retrieval of both event and profile memory is effectively solved (hit@10 = 1.000), so answer quality is governed by generation. The larger E4B model produces a statistically reliable improvement on personalized-semantic answering and a directional, non-significant edge on episodic recall, so the model-size payoff is specific to personalization rather than general. Personalized answering stays hard for both models in absolute terms, and reasoning recovers the hardest cases, most for E4B. The work that follows closes the loop back to the original motivation: add the productivity-assistant demo as a qualitative bridge, run the full none/episodic/semantic/combined comparison the pitch described, scale LaMP-QA past the 100-question sample, and test a full-distribution reasoning condition instead of only the hard tail.

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Appendix A
Project Plan Summary

This appendix updates the pitch's 70-hour plan to show completed, changed, removed, and remaining work. The hours are a reasoned reconstruction of effort from the project's commit history, not clocked time; the total is held at the pitch's approximately 70-hour estimate. The scope narrowed from a synthetic productivity assistant with four memory conditions to a benchmark-first evaluation of retrieved episodic and semantic memory on small local models. Two commits mark the change: "personalization pipeline *without generated dataset*" (drop synthetic data) and "*No QASPER*" (cull the third benchmark). Most planned tasks are complete; the remaining work is the presentation, its recording, and confirming repository access.

TABLE I
Updated Project Plan Summary

Phase	Work	Status	Hrs
Literature review	Prior-art survey; benchmark and method sourcing; Zotero curation	Done	12
Data normalization	EPBench and LaMP-QA normalization, sampling, validation	Done	12
Model/server setup	Inference models; two local `llama-server` instances; retrieval	Done	9
Evaluation runs	EPBench and LaMP-QA generation runs (E2B, E4B)	Done	12
Grading/judging	Deterministic EPBench grader; LaMP-QA rubric judge	Done	9
Analysis	Metric suite; bootstrap CIs; bottom-10 sub-eval; figures	Done	8
Synthetic dataset	Planned generated dataset	Changed	2
QASPER benchmark	Exploratory third benchmark	Removed	n/a
Productivity UI	Interactive assistant	Deferred	n/a
Reporting	Outline, figures, this report	In prog.	4
Presentation	Slides, 5-minute recording	Remaining	n/a
Repo access check	Confirm staff access to private repo	Remaining	2
		Total Estimated Hours	**70**

Appendix B
Code Repository

The full pipeline, meaning normalization, validation, evaluation runners, graders, the figure-generation script, and all evaluation artifacts, lives in the project repository on the course GitHub organization:

https://github.gatech.edu/OMSCS-CS6795-Summer-2026/episodic-semantic-memory-assistant

The repository is private per the assignment's requirement; course staff will be granted viewing access before the submission deadline. No CITI training certificate is included because the project involves no human subjects.